The empirical recurrence for OEIS A229667, proved by counting defective colourings

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Abstract

OEIS A229667 counts the colourings of an array with $K = 4$ colours that are proper except at exactly two adjacent pairs, the colours being introduced in row major order. Neither feature is local: the number of monochromatic adjacent pairs is a property of the whole array, and the row major clause fixes a representative of each relabelling class. Both are removed by machinery already in hand — the running count of mistakes goes into the state, capped at 2, and the colour clause disappears into the identity $4!\,a(n) = \sum_i \binom{4}{i} D_{4-i} L_i(n)$ with D_m the derangement numbers. What is left is a walk count on $S = 300$ vertices, and the empirical recurrence of order 6 contributed by R. H. Hardin follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A229667 is "Number of defective 4-colorings of an nX3 0..3 array connected horizontally, antidiagonally and vertically with exactly two mistakes, and colors introduced in row-major 0..3 order.". It has offset 1 and begins

$$1, 61, 1555, 19805, 194575, 1673561, 13271403, 99602405, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 21*a(n-1) -171*a(n-2) +679*a(n-3) -1368*a(n-4) +1344*a(n-5) -512*a(n-6) for n>7.

As of the "Last modified" line on the live entry (Jul 23 2025 05:45:59, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \ = \ 21\,a(n-1) - 171\,a(n-2) + 679\,a(n-3) - 1368\,a(n-4) + 1344\,a(n-5) - 512\,a(n-6) \quad (1)$$

for all $n > 7$.

2 Mistakes, and the colour clause

The arrays have 3 columns and $L = n$ rows, entries in $\{0, \ldots, 4-1\}$. Two cells are adjacent when they differ by one of

$$\mathcal{N} = \{(0,1),\ (1,-1),\ (1,0)\}$$

(as (row shift, column shift)); each unordered adjacency is represented by exactly one offset, and every offset has a nonnegative row shift, so the pairs inside a row are counted when that row is laid down and the pairs joining two rows when the later one is. Call an adjacent pair a *mistake* when $x_{t,u} = x_{t+d_1,\,u+d_2}$. The entry asks for exactly $E = 2$ mistakes.

That is a statement about the whole array, so no bounded window sees it; but the running number of mistakes can be carried in the state, capped at 2, the states past 2 being absent because a count that has passed the budget can never come back.

The condition is stated in equalities, so it is invariant under relabelling the colours, and so is the number of mistakes. "Colors introduced in row-major order" keeps exactly one colouring from each relabelling class, so the entry counts equality PATTERNS with at most $K = 4$ classes.

Lemma 1. *Let $L_i(n)$ count the colourings over i colours with exactly 2 mistakes, and $N_j(n)$ the admissible patterns with exactly j classes. Then $L_i = \sum_j N_j\, i(i-1)\cdots(i-j+1)$ and*

$$4! \sum_{j \le 4} N_j(n) = \sum_{i=0}^{4} \binom{4}{i} D_{4-i} L_i(n), \qquad D_m = m! \sum_{t=0}^{m} \frac{(-1)^t}{t!}.$$

Proof. A colouring is a pattern together with an injection of its classes into the palette, and a pattern with j classes admits $i(i-1)\cdots(i-j+1)$ of them, which is the first identity. Writing the falling factorial as $j!\binom{i}{j}$ and inverting binomially gives $j! N_j = \sum_i (-1)^{j-i}\binom{j}{i} L_i$; summing over $j \le 4$ and exchanging the order of summation makes the coefficient of L_i equal to $\sum_{t \le 4-i} (-1)^t/(i!\,t!)$, which multiplied by 4! is $\binom{4}{i} D_{4-i}$, an integer. □

3 Each L_i is a walk count

Fix i and take as vertices the pairs (r, c) with $r \in \{0, \ldots, i-1\}^3$ a row and $0 \le c \le 2$ a running count. Write $\beta(r, s)$ for the number of mistakes joining rows r and s plus the number of mistakes inside s. Declare $(r, c) \to (s, c + \beta(r, s))$ an edge when $c + \beta(r, s) \le 2$, let $\mathbf{s}_i$ mark the vertices whose count is the number of mistakes inside r alone, and $\mathbf{e}_i$ those with $c = 2$.

Lemma 2. *$L_i(n) = \mathbf{s}_i^\top P_i^{L-1} \mathbf{e}_i$ for $L \ge 1$, where P_i is the adjacency matrix of that digraph.*

Proof. A walk of length $L-1$ determines the sequence of L rows, and conversely; by induction the count component after t rows is the number of mistakes among the first t rows, since each step adds exactly the mistakes joining rows t and $t+1$ together with those inside row $t+1$, and each unordered pair is named by exactly one offset. The walk exists precisely when the running count never exceeds 2, which for a nondecreasing count follows from its final value being 2, and $\mathbf{e}_i$ imposes that final value. The state carries a row, not a boundary, so L rows make a walk of length $L-1$. □

Assembling the 4 chains block-diagonally into N of size $S = 300$ with the weights of Lemma 1 in the start vector $\mathbf{v}$ and the end markers in $\mathbf{u}$,

$$4!\, a(n) \;=\; \mathbf{v}^\top N^{L-1} \mathbf{u}, \qquad L = n. \tag{2}$$

4 The criterion, and the computation

Let $q(t) = t^6 - \sum_i c_i\, t^{6-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 3. *With n_0 the least index for which $L \ge 1$, $o = 1n_0 + 0 - 1$ and $h_j = G^j N^o \mathbf{u}$, we have for $j \ge 6$*

$$4!\Big(a(n_0 + j) - \sum_i c_i\, a(n_0 + j - i)\Big) = \mathbf{v}^\top G^{j-6} w, \qquad w = h_6 - \sum_i c_i h_{6-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for all $m \ge 0$, and $m < S$ suffices.

Proof. Substitute (2) and factor; 4! is a nonzero scalar. Cayley–Hamilton makes G^S an integer combination of $I, G, \ldots, G^{S-1}$, so every later value repeats that combination of earlier ones. $\square$

Theorem 4. *The recurrence* (1) *holds for every* $n > 7$*, which is the statement of Section 1.*

Proof. The digraph was built directly from the definitions above. The vector w was formed by 6 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \ldots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 7+1$ onwards, and the run continued until $S + 1$ consecutive zeros had been seen. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the right-hand side of (2), divided by 4!, was evaluated at every one of the 19 published indices and agrees with the entry's DATA exactly, the quotient coming out an integer each time; no shift was fitted, the number of rows being $L = n$ from the entry's own name and the offset saying which index carries the first published term. Double-counting an unordered pair, or mistaking the budget, changes the counts at once.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 3 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix.

References

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